

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

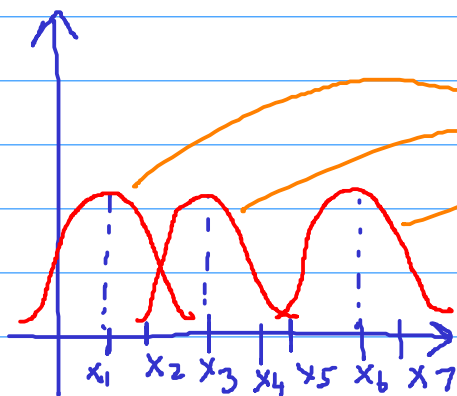
$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \times n \times \sigma^2 = \sigma^2/n$$

$$E(\bar{X}) = \frac{1}{n} \times n \mu = \mu$$

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

$$\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \sim N(0, 1)$$

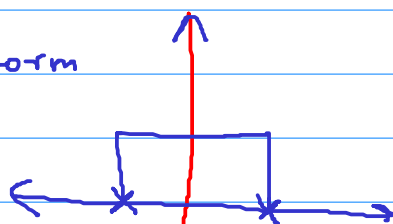


Kernel density estimator
= bandwidth / smoothing parameter of the KDE

$$\hat{f}_n(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) \quad h \text{ small} \rightarrow$$

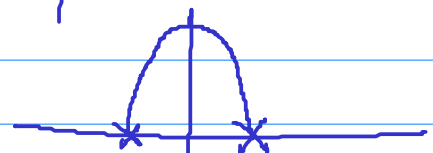
$$= \frac{1}{n} \sum_{i=1}^n \frac{e^{-\frac{(x - x_i)^2}{2h^2}}}{h\sqrt{2\pi}} \quad \checkmark$$

Uniform



$$K(x) = \begin{cases} 1 & -h \leq x \leq h \\ 0 & \text{otherwise} \end{cases}$$

Epanechnikov



$$K(x) = \frac{3}{4} \max(h - x^2, 0)$$

Kernel ① smooth, diff. function

② symmetric about 0

③ $\lim_{x \rightarrow -\infty} K(x) = 0 = \lim_{x \rightarrow +\infty} K(x)$

Bias of the KDE

point at which the PDF is evaluated

samples

$$E(\hat{p}_n(x_0)) = E\left[\frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x_0}{h}\right)\right]$$

$= \frac{1}{nh} \times n E\left(K\left(\frac{x_i - x_0}{h}\right)\right) \because \text{all } X_i\text{'s are i.i.d. distr.}$

$$= \frac{1}{h} \int K\left(\frac{x - x_0}{h}\right) p(x) dx$$

$$= \frac{1}{h} \int K(y) p(hy + x_0) dy \quad \begin{matrix} y = \frac{x - x_0}{h} \\ dy = dx/h \end{matrix}$$

for small h

$$p(x_0 + hy) = p(x_0) + hy p'(x_0) + \frac{1}{2} h^2 y^2 p''(x_0) + o(h^2)$$

eg h^3, h^4 (growing slower than h^2)

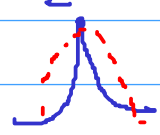
$$= \int K(y) \left[p(x_0) + hy p'(x_0) + \frac{h^2}{2} y^2 p''(x_0) + o(h^2) \right] dy$$

$$= p(x_0) \int K(y) dy + h p'(x_0) \int y K(y) dy + \frac{h^2}{2} p''(x_0) \int y^2 K(y) dy + o(h^2) \int K(y) dy$$

μ_{2K}

$$= p(x_0) + \frac{h^2 p''(x_0)}{2} \mu_{2K} + O(h^2)$$

$x_0^* \in [x_0, x_0 + h]$

$$\text{bias}(\hat{p}_n(x_0)) = h^2 \frac{p''(x_0)}{2} \mu_{2K} + O(h^2)$$


Taylor series

$$\textcircled{1} \quad f(x) = f(x_0) + h f'(x_0^*)$$

$x_0^* \in [x_0, x]$

$$\textcircled{2} \quad f(x) \doteq f(x_0) + h f'(x_0) + \boxed{h^2} + \dots$$

$$f(x) = f(x_0) + h f'(x_0) + \frac{h^2}{2} f''(x_0^*)$$

$$\text{Var}(\hat{p}_n(x_0)) = \text{Var}\left(\frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - x_0}{h}\right)\right)$$

$$= \frac{1}{n^2 h^2} n \text{Var}\left[K\left(\frac{x_i - x_0}{h}\right)\right]$$

$$= \frac{1}{nh^2} \left[E\left[K^2\left(\frac{x_i - x_0}{h}\right)\right] - E^2(K) \right]$$

$$\leq \frac{1}{nh^2} E\left[K^2\left(\frac{x_i - x_0}{h}\right)\right]$$

$\rightarrow \geq 0$

$$= \frac{1}{nh^2} \int K^2 \left(\frac{x - x_0}{h} \right) p(x) dx$$

$$= \frac{1}{nh^2} \int K^2(y) p(x_0 + hy) / h dy \quad y = \frac{x - x_0}{h}$$

$$= \frac{1}{nh} \int K^2(y) [p(x_0) + h y p'(x_0) + o(h)] dy$$

$$= \frac{1}{nh} \left[p(x_0) \int K^2(y) dy + h p(x_0) \int y K^2(y) dy + o(h) \int K^2(y) dy \right]$$

$$= \frac{p(x_0)}{nh} \sigma_{2K}^2 + h p'(x_0) \cdot 0 + \frac{o(h)}{nh} \sigma_{2K}^2$$

$$= \frac{p(x_0)}{nh} \sigma_{2K}^2$$

Bias-Variance tradeoff

As $h \uparrow$, variance $\downarrow$

$$\frac{h^5 = h^4}{nh} \quad \frac{o(h)}{nh}$$

$$MSE = bias^2 + variance$$

$$\leq \alpha h^4 + \frac{\beta}{nh} + \frac{1}{n}$$

$$\text{To minimize MSE, } \frac{\partial MSE}{\partial h} = 4\alpha h^3 + \frac{\beta}{n} \frac{(-1)}{h^2} = 0$$

$$\rightarrow h = O(n^{-1/5})$$

$$MSE \leq \alpha n^{-4/5} + \beta n^{-4/5} = O(n^{-4/5})$$

Cross-validation

$X_1 \dots X_n$

$X_1 \dots X_{n_1}$

$$n_1 = 90\% \cdot n$$

Training set

$X_{n_1+1} \dots X_n$

$$10\% \cdot n$$

Validation set

$$\hat{p}_{n_1}(x; h) = \frac{1}{n_1 h} \sum_{i=1}^{n_1} K\left(\frac{x - x_i}{h}\right)$$

Build a PDF using diff. values of h .